

DAY 00 / MASTER INDEX

Notes Collection

Topic Index

A guided map of the complete 11-day collection: production GenAI foundations, workflow integration, system design, core DSA patterns, and clean architecture.

11 TOPIC NOTES	425 PDF PAGES	24,056 SOURCE LINES
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RECOMMENDED PATH

1	Build the GenAI foundation Days 1-4: learn retrieval, data frameworks, application composition, and stateful workflows.
2	Connect and consolidate the stack Days 5-7: add MCP, understand component boundaries, and revise the complete production architecture.
3	Strengthen engineering depth Days 8-11: practice system design, reusable DSA patterns, and maintainable GenAI architecture.

HOW TO USE THIS INDEX	Choose a topic below, open the matching Markdown source for editing, or use the companion PDF in Notes/pdfs for reading and review.
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MODULE 01 / DAYS 01-04

GenAI foundations and orchestration

Start with the retrieval pattern, then move through data tooling, application composition, and explicit workflow control.

DAY

01

Vanilla RAG - End to End

Builds the retrieval-augmented generation mental model from first principles, then follows ingestion and query-time execution into production.

KEY COVERAGE documents and chunks; embeddings and vector search; keyword, semantic, and hybrid retrieval; reranking; grounding; citations; evaluation; cost and latency

SOURCE [Day-01-Vanilla-RAG-End-to-End.md](#) | **PDF** 51 pages

DAY

02

LlamaIndex - Foundations and Workflows

Explains how LlamaIndex organizes data ingestion, indexing, retrieval, response synthesis, workflows, and agents.

KEY COVERAGE documents and nodes; connectors and parsers; transformations; vector indexes and stores; retrievers; query engines; rerankers; metadata; optimization

SOURCE [Day-02-LlamaIndex-Foundations-and-Workflows.md](#) | **PDF** 34 pages

DAY

03

LangChain - End to End

Covers the application-composition layer for prompts, models, tools, retrieval, structured outputs, and reusable chains.

KEY COVERAGE models; prompt templates; few-shot prompting; output parsers; structured output; retrievers; tools; chains; memory and state; callbacks and observability

SOURCE [Day-03-LangChain-End-to-End.md](#) | **PDF** 27 pages

DAY

04

LangGraph - AI Workflows

Introduces explicit, stateful orchestration for multi-step AI workflows with recovery, approvals, and production controls.

KEY COVERAGE state; nodes and edges; conditional routing; tool execution; checkpointing; durable execution; human-in-the-loop; streaming; retries; bounded control flow

SOURCE [Day-04-LangGraph-AI-Workflows.md](#) | **PDF** 33 pages

MODULE 02 / DAYS 05-07

Integration, governance, and revision

Connect AI systems to external capabilities, understand how the layers fit together, and consolidate the full production stack.

DAY 05	MCP - End-to-End Overview Explains the Model Context Protocol as a governed connection layer between AI hosts and external capabilities. KEY COVERAGE host, client, and server; tools, resources, and prompts; JSON-RPC; transports; stateful sessions; authentication; authorization; least privilege; governance SOURCE Day-05-MCP-End-to-End-Overview.md PDF 38 pages
DAY 06	How the Production AI Stack Fits Together Maps Vanilla RAG, LlamaIndex, LangChain, LangGraph, and MCP to distinct architectural roles in one end-to-end system. KEY COVERAGE category map; tool-selection boundaries; complete request lifecycle; framework overlap; vector databases and providers; security; observability; evaluation SOURCE Day-06-Interrelationship-of-RAG.md PDF 40 pages
DAY 07	Revision - The Complete Production AI Stack Consolidates Days 1-6 into a single revision guide with an integrated architecture, trade-offs, misconceptions, and interview framing. KEY COVERAGE RAG revision; LlamaIndex; LangChain; LangGraph; MCP; production controls; optimization; misconceptions; staff-level explanations; final memory map SOURCE Day-07-Revision-The-Complete-Production-AI-Stack.md PDF 51 pages
ARCHITECTURE MEMORY LINE	RAG retrieves evidence. LlamaIndex manages data-centric retrieval. LangChain composes application components. LangGraph controls stateful execution. MCP standardizes governed access to external capabilities.

MODULE 03 / DAYS 08-11

Engineering interviews and architecture

Translate the GenAI foundation into lead-level system design, coding-pattern fluency, and clean implementation boundaries.

DAY

08

System Design and LLD Interview Playbook

A lead-engineer framework for turning ambiguous requirements into estimates, APIs, data models, architecture, and a clear interview narrative.

KEY COVERAGE six-phase design framework; capacity estimation; API contracts; idempotency; data modeling; LLD; AgentRun case study; reliability; security; observability; trade-offs

SOURCE Day-08-System-Design-Interview-Prep.md | **PDF** 40 pages

DAY

09

DSA Core I - Arrays, Strings, Hashing and Prefix Sums

Builds the first set of reusable interview patterns, emphasizing recognition, complexity, Python templates, and edge cases.

KEY COVERAGE complexity; traversal; subarrays and substrings; prefix sums; dictionaries and sets; frequency counting; two-sum; duplicates; anagrams; prefix sum plus hashing

SOURCE Day-09-DSA-Core-I-Patterns.md | **PDF** 30 pages

DAY

10

DSA Core II - Two Pointers, Windows, Stacks and Queues

Extends the pattern library with movement, window, and ordering techniques used in coding interviews and practical systems.

KEY COVERAGE two pointers; fixed and variable sliding windows; longest substring; shortest subarray; stacks; balanced parentheses; queues; monotonic stacks; next greater element

SOURCE Day-10-DSA-Core-II-Patterns.md | **PDF** 31 pages

DAY

11

Clean Architecture and Design Patterns for GenAI

Applies clean architecture, SOLID, and classic design patterns to maintainable RAG and agent systems.

KEY COVERAGE dependency rule; SOLID; Factory; Strategy; Adapter; Decorator; Facade; layered architecture; provider abstractions; project structure; testing; trade-offs

SOURCE Day-11-GenAI-System-Architecture.md | **PDF** 50 pages

QUICK LOCATOR

Find a topic fast

Use this cross-reference when you know the concept you need but not the day that covers it.

Need to find...	Start with	Best note
RAG ingestion, chunking, embeddings	Core retrieval pipeline	Day 01
Indexes, nodes, query engines	Data-centric RAG tooling	Day 02
Prompts, chains, tools, output parsing	Application composition	Day 03
State, routing, checkpoints, approvals	Workflow orchestration	Day 04
MCP tools, resources, security	External capability protocol	Day 05
Which framework belongs where	Architecture boundaries	Days 06-07
Capacity, APIs, data model, LLD	System design interview	Day 08
Hashing, prefix sums, two-sum	DSA pattern recognition I	Day 09
Windows, stacks, queues	DSA pattern recognition II	Day 10
SOLID, patterns, clean architecture	Maintainable GenAI design	Day 11

COLLECTION SUMMARY

Source format	11 Markdown files in Notes/
Reading format	11 companion PDFs in Notes/pdfs/
Total coverage	24,056 source lines and 425 PDF pages
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